



Credit risk I. PD estimation

Risk Management in Financial Institutions

Lecture 8

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Market vs Credit Risk

- **Market risk** is the present and/or future **risk of losses** on on-balance sheet and off-balance sheet **positions** due to **changes in market prices** (changes in the price of bonds, securities, commodities, currencies or interest rates affecting the positions).
- **Credit risk** is the risk of loss to the profitability and capital position **arising from the non-performance (or nonperformance according to contractual terms) of counterparties**, i.e. the risk of **loss** resulting from partial or total failure to meet obligations (on or off balance sheet) to the financial institution.

Source: MNB ICAAP handbook

(<https://www.mnb.hu/letoltes/icaap-ilaap-bma-kezikonyv-2021januar.docx>)

Main credit risk subtypes

- **credit default risk:** the credit risk associated with certain financial and ancillary financial services or investment services activities (e.g. **lending and borrowing**, financial leasing, guarantees and sureties, other banking commitments, debt purchase);
- **counterparty credit risk:** the potential loss arising from the **failure** of a counterparty to a given (typically **derivative**) transaction **to meet its contractual obligations** before the transaction is closed (final settlement of cash flows).
- **counterparty credit risk (issuer risk):** the credit risk to the issuers of debt securities and equity securities, which includes the risk that the value of the security will decline because of a deterioration in the issuer's credit quality;
- **settlement risk:** the risk that a settlement through a payment and clearing and securities settlement system will not be completed as expected (even though the financial institution party fulfils its contractual obligation, it does not receive the consideration for it).

Source: Recommendation No 14/2021 (IX.16.) of the Magyar Nemzeti Bank on the commitment, measurement, management and control of credit risk (<https://www.mnb.hu/letoltes/14-2021-hitelkocka-ajanlas.pdf>) and MNB ICAAP Handbook (counterparty risk)

Credit risk concepts

- **Default event:** the counterparty to a financial contract (e.g. loan contract, security (bond), derivative) fails to perform as contractually agreed
 - Security, derivative: fails to pay exactly within the given time limit (e.g. fails to pay interest or principal on a given day or after a given day + 30 days grace period)
 - Credit agreement: 'looser' than this, see Basel definitions (e.g. 90 days without payment)
- **Default risk->probability of default (PD, %):**
 - a risk factor for credit risk
 - the probability that a default event will occur for a given counterparty/contract within a given time period (e.g. one year)
- **Loss given Default (LGD, %)=1-Recovery**
 - Another risk factor for credit risk
 - Amount/loss remaining after the cash inflows (Recovery) after the default event
 - Given as a % of EAD; often set as a fixed value (not a variable) for simplicity
- **Exposure (at Default, EAD):** exposure, the total amount of claims (expected to be) outstanding at the time of default
 - The third risk factor for credit risk
 - In some cases it is a "given" that does not need to be modelled, e.g. for loans, the outstanding principal
 - Derivatives - to be modelled - positive expected market value at default
- **Expected Credit Loss (ECL or EL)=PD*LGD*EAD**

Credit risk applications

- Theoretical pricing of risky bonds
 - (a) estimating risk-neutral **default probabilities**
 - (b) calculating the expected payoff from the instrument
 - (c) discounting the expected payoff at the risk-free rate
- Credit VaR – several methods
- Basel II IRB formula

$$\sum_i EAD_i \cdot LGD_i \cdot WCDR_i$$
$$WCDR_i = N \left[\frac{N^{-1}(PD_i) + \sqrt{\rho} N^{-1}(0,999)}{\sqrt{1 - \rho}} \right]$$

Estimating Default Probabilities

- Alternatives:
 - Use historical data
 - Use credit spreads (CDS spread, bond yield spread)
 - Use Merton's model

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Historical Data

Historical data provided by rating agencies can be used to estimate the probability of default

Ratings change slowly but steadily as default risk changes: example of a major bank and their evolution over time

Deutsche Bank AG [DE] | <https://www.db.com/ir/en/current-ratings.htm>

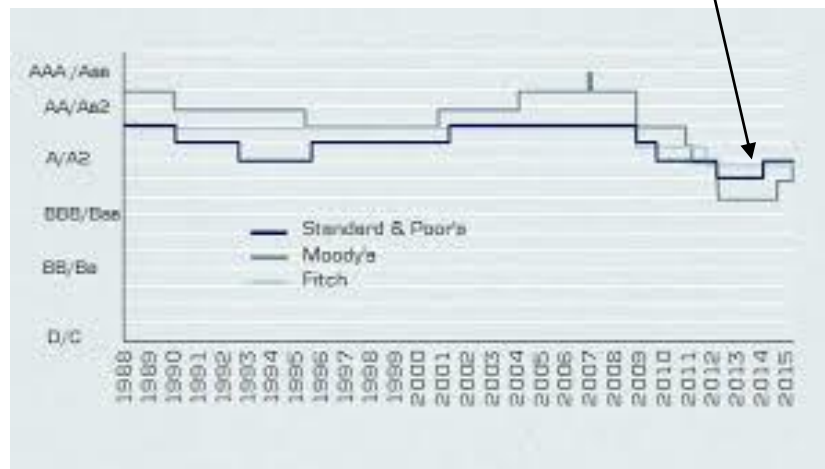
Ratings

Ratings Deutsche Bank AG

Moody's Investors Service

	Rating	Last rating change	Outlook
Counterparty Risk Assessment	A3 (cr)	May 23, 2016	Stable
Long-term deposit rating	A3	May 23, 2016	Stable
Short-term deposit rating	P-2	May 23, 2016	Stable
Long-term senior debt rating	Baa2	May 23, 2016	Stable
Short-term senior debt rating	P-2	May 23, 2016	Stable
Baseline Credit Assessment	ba1	May 23, 2016	Stable
Non-deferrable dated subordinated (Tier 2)	Ba2	May 23, 2016	Stable
Legacy Tier 1 (Basel 2.5)	B1	May 23, 2016	Stable
Additional Tier 1 (Basel 3)	B1	May 23, 2016	Stable
Pfandbrief	Aaa	–	Stable

Effects of the financial crisis



Cumulative Average Default Rates % (1970-2013, Moody's) Table 19.1, page 402

	Time (years)						
	1	2	3	4	5	7	10
Aaa	0.000	0.013	0.013	0.037	0.104	0.241	0.489
Aa	0.022	0.068	0.136	0.260	0.410	0.682	1.017
A	0.062	0.199	0.434	0.679	0.958	1.615	2.759
Baa	0.174	0.504	0.906	1.373	1.862	2.872	4.623
Ba	1.110	3.071	5.371	7.839	10.065	13.911	19.323
B	3.904	9.274	14.723	19.509	23.869	31.774	40.560
Caa	15.894	27.003	35.800	42.796	48.828	56.878	66.212

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Interpretation

- The table shows the probability of default for companies starting with a particular credit rating
- A company with an initial credit rating of Baa has a probability of 0.174% of defaulting by the end of the first year, 0.504% by the end of the second year, and so on

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Do Default Probabilities Increase with Time?

- For a company that starts with a good credit rating default probabilities tend to increase with time
- For a company that starts with a poor credit rating default probabilities tend to decrease with time

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Hazard Rate vs. Unconditional Default Probability

- The hazard rate or default intensity is the probability of default over a short period of time conditional on no earlier default
- The unconditional default probability is the probability of default as seen at time zero

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Recovery Rate

The recovery rate for a bond is usually defined as the price of the bond 30 days after default as a percent of its face value

$$\text{LGD} = 1 - \text{Recovery rate}$$

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Recovery Rates; Moody's: 1982 to 2013

(Table 19.2, page 404)

Class	Ave Rec Rate (%)	Ave LGD (%)
Senior secured bond	52.2	47.8
Senior unsecured bond	37.2	62.8
Senior subordinated bond	31.0	69.0
Subordinated bond	31.4	68.6
Junior subordinated bond	24.7	75.3

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Estimating Default Probabilities

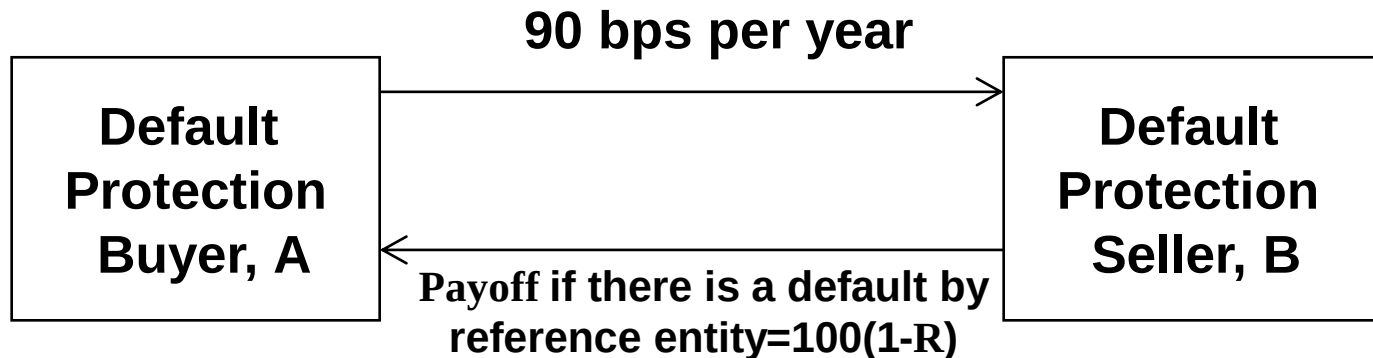
- Alternatives:
 - Use historical data
 - Use credit spreads – CDS OR Bond Yields
 - Use Merton's model

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Credit Default Swaps

- Buyer of the instrument acquires protection from the seller against a default by a particular company or country (the reference entity)
- Example: Buyer pays a premium of 90 bps per year for \$100 million of 5-year protection against company X
- Premium is known as the *credit default spread*. It is paid for life of contract or until default
- If there is a default, the buyer has the right to sell bonds with a face value of \$100 million issued by company X for \$100 million (Several bonds may be deliverable)

CDS Structure (Figure 19.1, page 406)



Recovery rate, R , is the ratio of the value of the bond issued by reference entity immediately after default to the face value of the bond

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Using CDS Prices to Predict Default Probabilities

Average hazard rate between time zero and time t is to a good approximation
$$\bar{\lambda} = \frac{s(t)}{1 - R}$$

where $s(t)$ is the credit spread calculated for a maturity of t and R is the recovery rate (1-LGD)

In other words, this is an **estimate of the average** probability of a default **per year** over the total period, conditional on no earlier default

For a 1-year CDS:

CDS spread(1yr) = PD(1yr) * LGD,

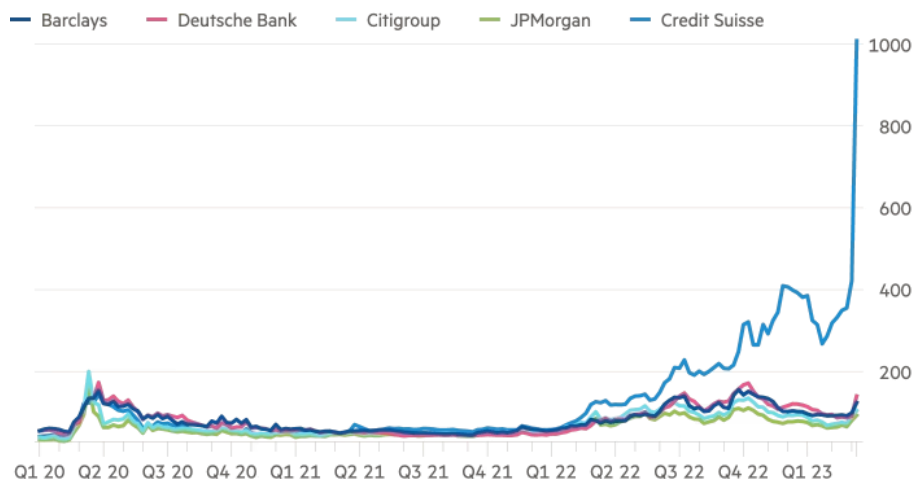
and rearranging gives

PD(1yr) = CDS spread(1yr) / LGD

Credit Suisse CDS price – showing „crisis” periods

Credit Suisse's CDS prices have soared far above other banks'

Five-year US dollar CDS prices (basis points)



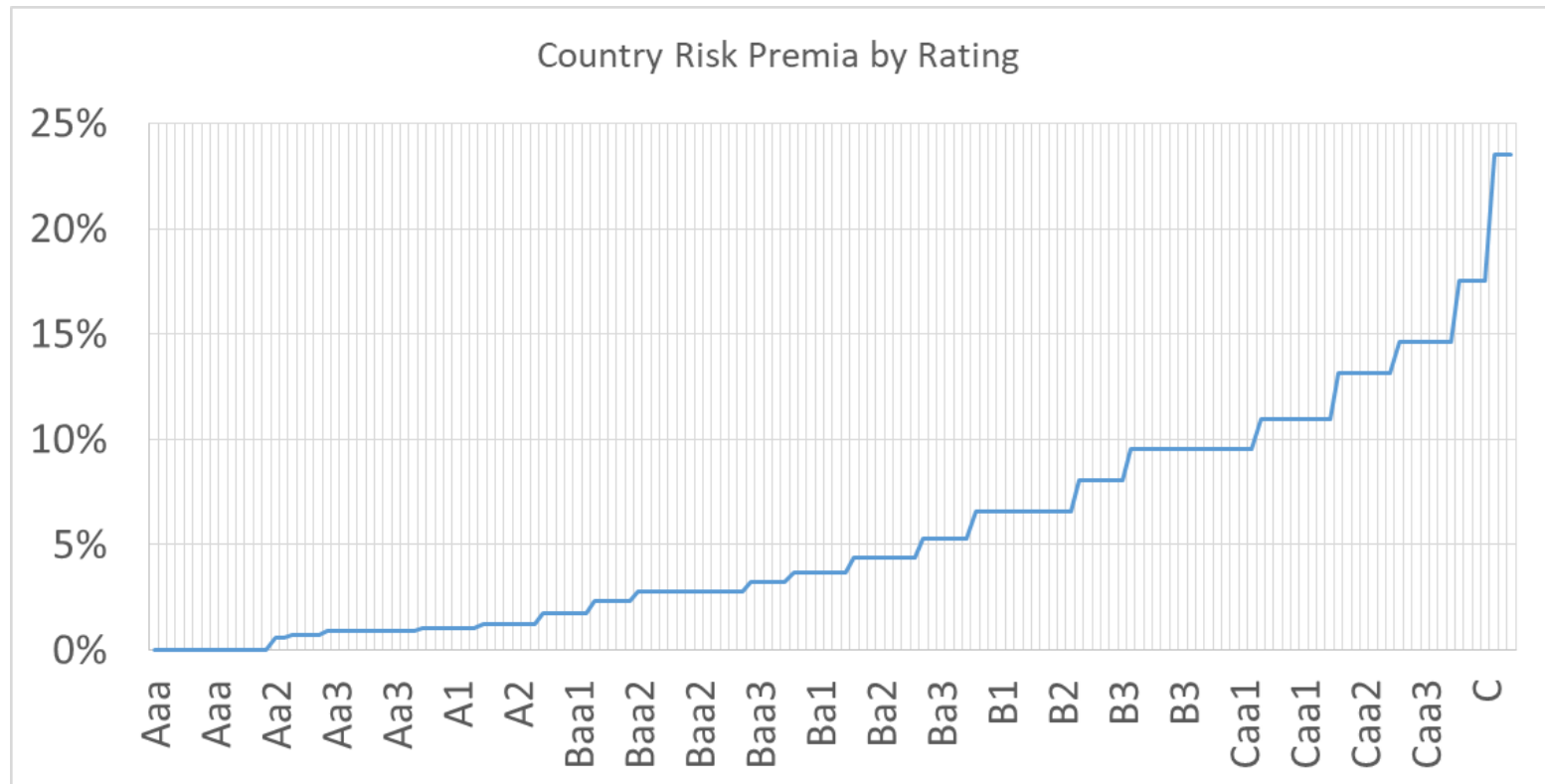
Source: Refinitiv
© FT



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BOND YIELDS

THEY INCREASE AS CREDIT RISK INCREASES



Source: Damodaran,

https://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/ctryprem.html

BEST CREDIT RATINGS: A FEW EXAMPL

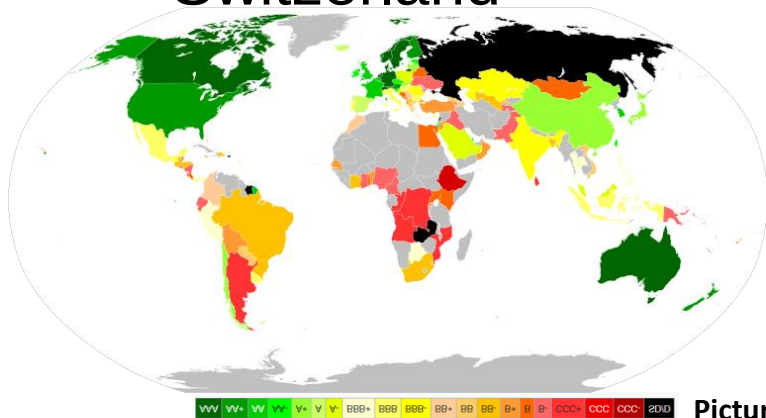
AAA countries:

Australia

Germany

Luxembourg

Switzerland



AAA corporations, entitites:

Johnson & Johnson

Microsoft

World Bank

EIB

Johnson & Johnson

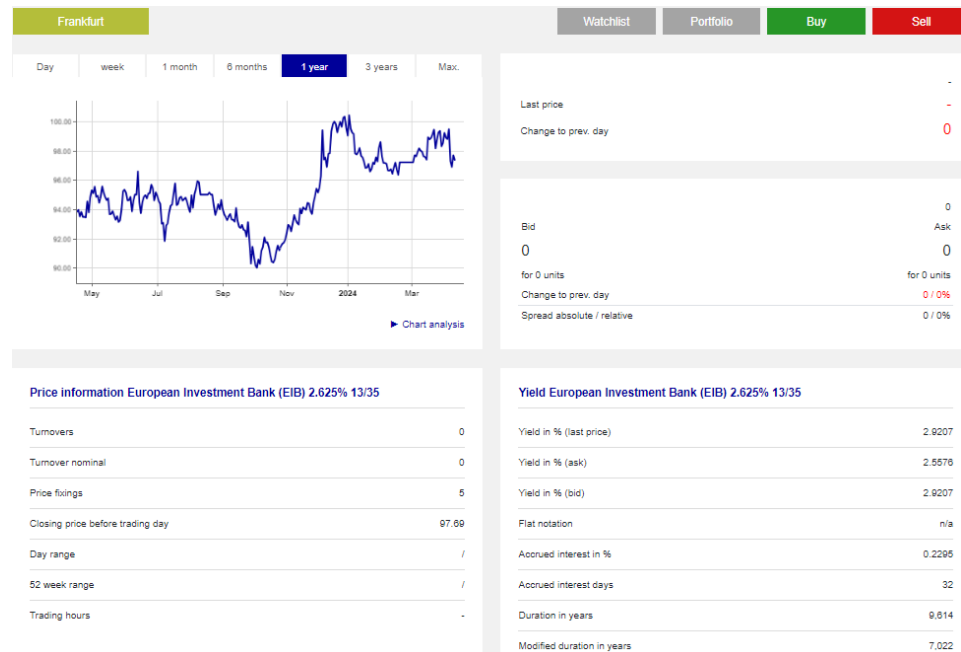


Picture Source: https://commons.wikimedia.org/wiki/File:BlankMap-World_S%26P.svg

CREDIT SPREAD: DIFFERENCE IN YIELDS BETWEEN AAA AND HIGHER RISK BONDS

European Bond Yields and Spreads

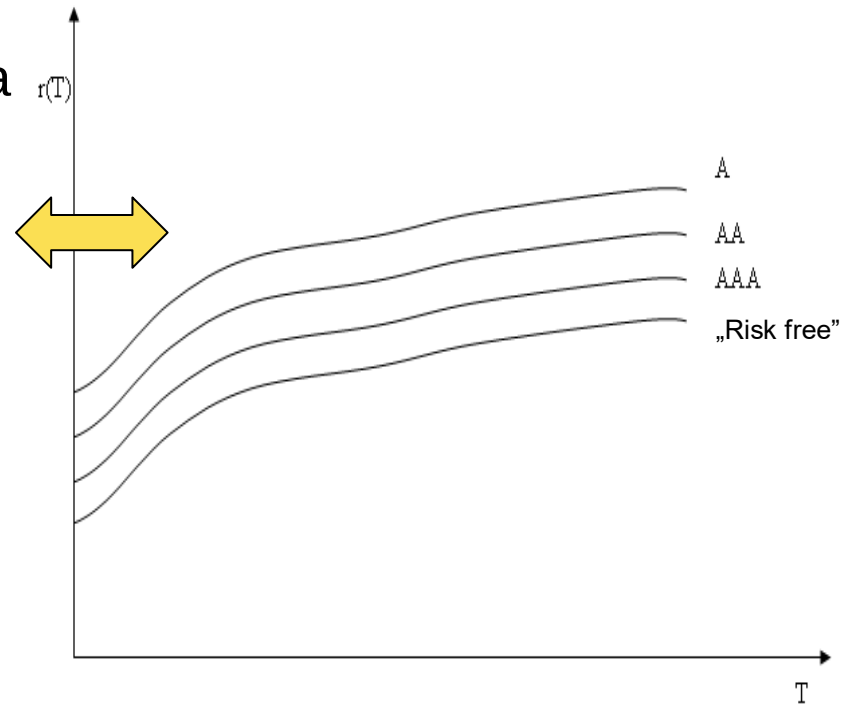
	2yr	5yr	10yr	30yr
Country	Yield	Spread*	Close†	Change‡
Germany (2.20% 15 Feb 2034)	2.41	-	2.34	0.08
EU (3.25% 04 Jul 2034)	2.98	+57	2.91	0.07
France (3.5% 25 Nov 2033)	2.92	+50	2.84	0.08
Belgium (2.85% 22 Oct 2034)	2.97	+56	2.89	0.08
Italy (4.2% 01 Mar 2034)	3.77	+136	3.71	0.06
Spain (3.25% 30 Apr 2034)	3.23	+82	3.16	0.07
Denmark (2.25% 15 Nov 2033)	2.45	+3	2.37	0.08
Finland (3.0% 15 Sep 2033)	2.86	+45	2.79	0.07
Netherlands (2.5% 15 Jul 2034)	2.71	+29	2.63	0.08
Austria (2.90% 20 Feb 2034)	2.93	+51	2.85	0.08
Czech Republic (4.90% 14 Apr 2034)	4.35	+193	4.23	0.12
Ireland (2.6% 18 Oct 2034)	2.83	+42	2.76	0.07
Slovenia (3.0% 10 Mar 2034)	3.17	+76	3.08	0.09
Hungary (4.75% 24 Nov 2032)	7.08	+467	6.95	0.13
Portugal (2.875% 20 Oct 2034)	3.08	+67	3.01	0.07
Slovakia (3.625% 8 Jun 2033)	3.53	+112	3.45	0.08



Sources: <https://www.mtsmarkets.com/european-bond-spreads>
<https://www.boerse-frankfurt.de/bond/xs0878008225-european-investment-bank-eib-2-625-13-35>

Credit Default Swaps and Bond Yields (page 409-410)

- Portfolio consisting of a 5-year par yield corporate bond that provides a yield of 6% and a long position in a 5-year CDS costing 100 basis points per year is (approximately) a long position in a riskless instrument paying 5% per year
- What are arbitrage opportunities in this situation if risk-free rate is 4.5%? What if it is 5.5%?



CREDIT SPREAD DIFFERS ALONG THE YIELD CURVE

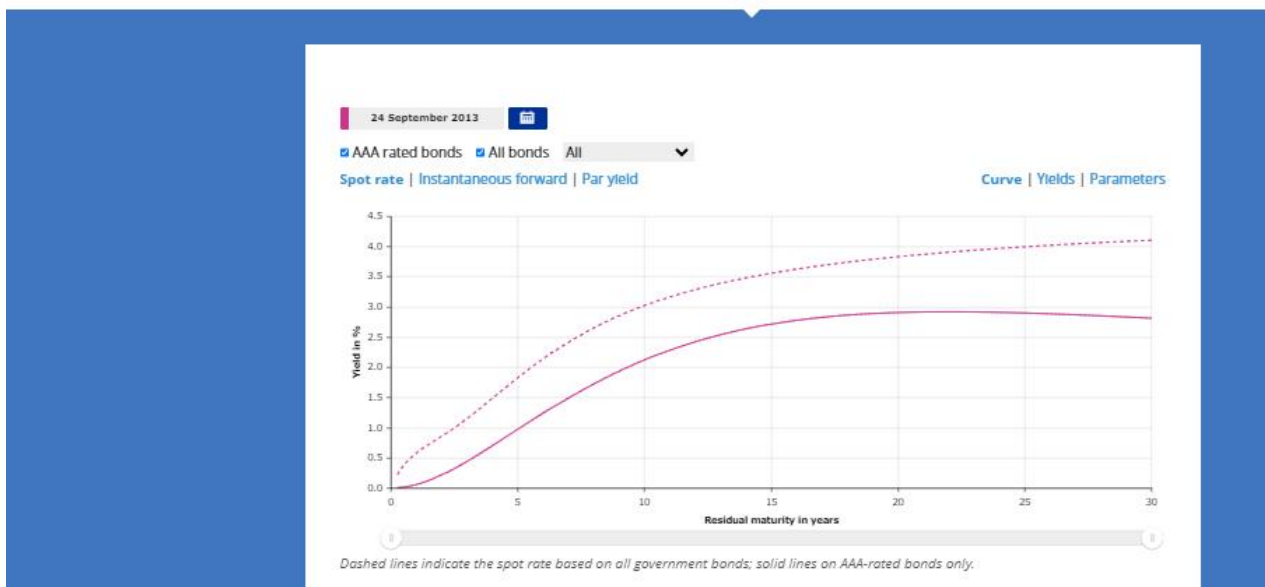


Euro area yield curves

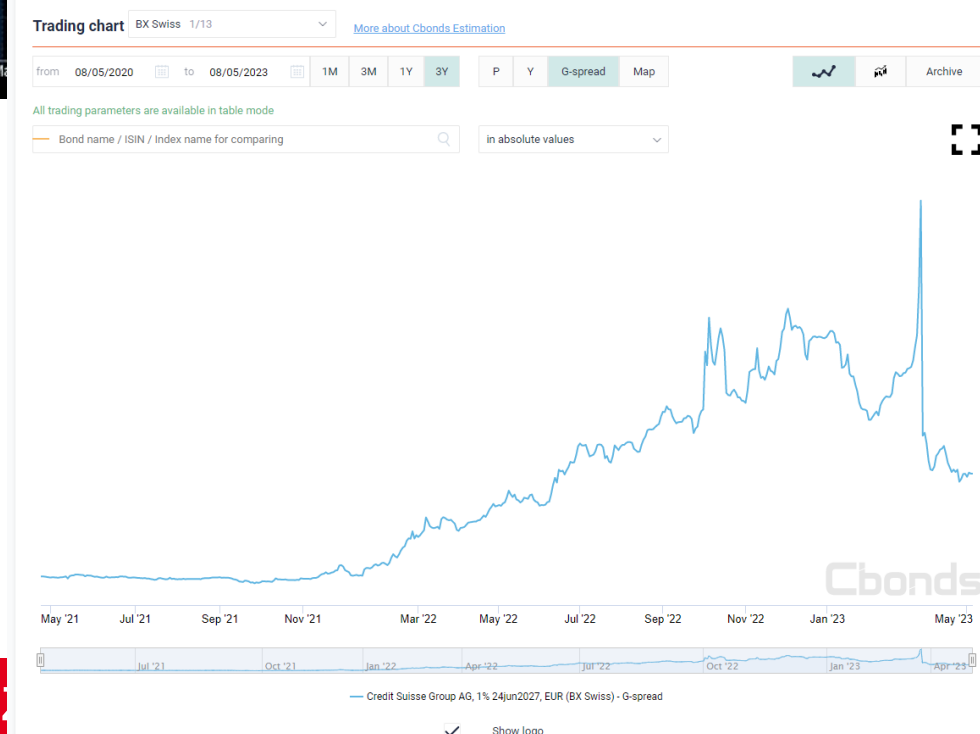
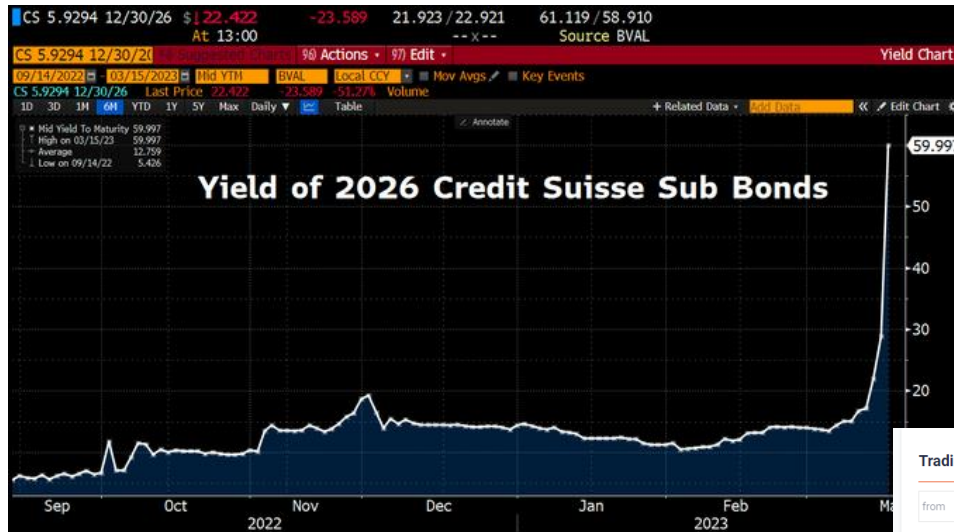
A yield curve is a representation of the relationship between market remuneration rates and the remaining time to maturity of debt securities. A yield curve can also be described as the term structure of interest rates. The ECB publishes several yield curves, as shown below.

It is updated every TARGET business day at noon (12:00 CET). No data or other information are provided regarding any day on which the relevant trading venue from which the euro area yield curve data are sourced is not open for business.

https://www.ecb.europa.eu/stats/financial_markets_and_interest_rates/euro_area_yield_curves/html/index.en.html



Credit Suisse bond yield and spread – indicating the jump in default probability in March 2023



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**Thank you for your
attention!**

